THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 4: **Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $(x-2)^2+1=2x-3$	(c) gives $x^2-6x+8=0$; others reduce to linear
2	**(a)** $4, -3$	$(x-4)(x+3)=0$
3	**(a)** 0	$D=4-4(3)(\frac{1}{3})=0$
4	**(c)** no real roots	$D=1-4=-3<0$
5	**(b)** Incorrect; $D=0$, so the roots are equal $(3,3)$	$D=36-36=0$
6	**(a)** 1	$k-3+2=0$
7	**(b)** ± 6	$k^2-36=0$
8	**(b)** Student B is correct	$x(x-5)=0$; dividing by x loses the root 0
9	**(a)** $\frac{-b\pm\sqrt{b^2-4ac}}{2a}$	Quadratic formula
10	**(a)** $x^2+x-306=0$	$x(x+1)=306$

SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2x^2+4x-3x-6=(2x-3)(x+2)=0 \Rightarrow x=\frac{3}{2}, -2$	1+1
12	$D=36-24=12>0 \Rightarrow$ two distinct real roots	1+1
13	$k^2-24=0 \Rightarrow k=\pm 2\sqrt{6}$	1+1
14	$\sqrt{2}x^2+5x+2x+5\sqrt{2}=(x+\sqrt{2})(\sqrt{2}x+5)=0 \Rightarrow x=-\sqrt{2}, -\frac{5}{\sqrt{2}}$. The student is **correct**.	1+1

SECTION — C (Short Answer) — Answers 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$D=49-24=25$; $x=\frac{7\pm 5}{4} \Rightarrow 3, \frac{1}{2}$	1+1+1
16	$x(27-x)=182 \Rightarrow x^2-27x+182=0 \Rightarrow (x-13)(x-14)=0 \Rightarrow$ 13 and 14	1+1+1
17	$x(2x+1)=528 \Rightarrow 2x^2+x-528=0$; $D=4225 \Rightarrow x=\frac{-1+65}{4}=16$. Breadth 16 m, length 33 m	1+1+1
	OR: $kx^2-2kx+6=0$; $D=4k^2-24k=0 \Rightarrow k=6$ $(k\neq 0)$	1+1+1
18	$l+b=40$, $lb=400 \Rightarrow x^2-40x+400=0$, $D=0 \Rightarrow x=20$. Yes: $20\text{ m}\times 20\text{ m}$	1+1+1
	OR: $(x-4)(16-x)=48 \Rightarrow x^2-20x+112=0$; $D=400-448<0 \Rightarrow$ **not possible**	1+1+1

SECTION — D (Long Answer) — Answers 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\frac{132}{x}-\frac{132}{x+11}=1 \Rightarrow x^2+11x-1452=0$; $D=5929=77^2 \Rightarrow x=33$. Passenger 33 km/h, express 44 km/h. Verify: $4-3=1$ h ✓	2+2+1
	OR: $\frac{1}{x-3}+\frac{1}{x+5}=\frac{1}{3} \Rightarrow x^2-4x-21=0 \Rightarrow (x-7)(x+3)=0 \Rightarrow$ 7 years. Verify: $\frac{1}{4}+\frac{1}{12}=\frac{1}{3}$ ✓	2+2+1
20	(i) $x(55-x)=750 \Rightarrow x^2-55x+750=0$ (ii) $(x-25)(x-30)=0 \Rightarrow$ 25 or 30 toys (iii) Both 25 (₹30 each) and 30 (₹25 each) satisfy the condition; the claim is incomplete.	1+3+1
	OR: $\frac{1}{x}+\frac{1}{x-10}=\frac{8}{75} \Rightarrow 4x^2-115x+375=0$; $D=7225=85^2 \Rightarrow x=25$ (or 3.75, rejected). Smaller 25 h, larger 15 h	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 4: **Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $(x+1)^2 = x(x+3)$	(c) reduces to $x - 1 = 0$ (linear)
2	**(a)** $1, \frac{3}{2}$	$(2x-3)(x-1) = 0$
3	**(a)** 49	$25 + 24 = 49$
4	**(c)** not real	$D < 0 \Rightarrow$ no real roots
5	**(b)** Incorrect; e.g. $x^2 + 1 = 0$ has no real root	A quadratic has at most two real roots
6	**(a)** 1	$4 + 2p - 6 = 0$
7	**(b)** $\pm\frac{4}{3}$	$9k^2 - 16 = 0$
8	**(a)** Student A is correct	$(x+3)(x-2) = 0$
9	**(a)** $x^2 + 3x - 10 = 0$	Sum -3, product -10
10	**(a)** $3x^2 - 10x + 3 = 0$	$x + \frac{1}{x} = \frac{10}{3}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(10x-1)^2 = 0 \Rightarrow x = \frac{1}{10}, \frac{1}{10}$	1+1
12	$D = 48 - 48 = 0 \Rightarrow$ equal real roots; $x = \frac{4\sqrt{3}}{6} = \frac{2}{\sqrt{3}}$ (both)	1+1
13	$4 - 4k = 0 \Rightarrow k = 1$; no real roots for $k > 1$	1+1
14	$9 - 6 - 15 = -12 \neq 0$, so 3 is not a root. $(x-5)(x+3) = 0 \Rightarrow 5, -3$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$D = 100 - 84 = 16$; $x = \frac{-10 \pm 4}{2\sqrt{3}} \Rightarrow -\sqrt{3}, -\frac{7}{\sqrt{3}}$	1+1+1
16	$x^2 + (x+2)^2 = 290 \Rightarrow x^2 + 2x - 143 = 0 \Rightarrow (x+13)(x-11) = 0 \Rightarrow$ 11 and 13	1+1+1
17	$x^2 + (x+30)^2 = (x+60)^2 \Rightarrow x^2 - 60x - 2700 = 0 \Rightarrow (x-90)(x+30) = 0 \Rightarrow$ 90 m and 120 m	1+1+1
	OR: $(k+1)^2 - 4(k+4) = 0 \Rightarrow k^2 - 2k - 15 = 0 \Rightarrow k = 5$ or -3	1+1+1
18	$2x^2 = 800 \Rightarrow x = 20$. Yes: breadth 20 m, length 40 m	1+1+1
	OR: $x(14-x) = 50 \Rightarrow x^2 - 14x + 50 = 0$; $D = 196 - 200 < 0 \Rightarrow$ **not possible**	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\frac{360}{x} - \frac{360}{x+5} = 1 \Rightarrow x^2 + 5x - 1800 = 0 \Rightarrow (x+45)(x-40) = 0 \Rightarrow$ 40 km/h. Verify: $9 - 8 = 1$ h ✓	2+2+1
	OR: $(x+2)(27-x) = 210 \Rightarrow x^2 - 25x + 156 = 0 \Rightarrow (x-12)(x-13) = 0 \Rightarrow$ (Maths, English) = (12, 18) or (13, 17)	2+2+1
20	(i) $(x+3)(x+29) = 360 \Rightarrow x^2 + 32x - 273 = 0$ (ii) $(x+39)(x-7) = 0 \Rightarrow$ 7 years (iii) 33 years; $10 \times 36 = 360$ ✓	1+3+1
	OR: Angle in a semicircle is $90°$: $x^2 + (x+7)^2 = 169 \Rightarrow x^2 + 7x - 60 = 0 \Rightarrow (x+12)(x-5) = 0 \Rightarrow x = 5$. Yes; 5 m and 12 m from the gates	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.